

DHANUSH INDLA

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with 5+ years of production experience building LLM-powered agents, RAG pipelines, and scalable data platforms for banking and healthcare. Delivered end-to-end AI systems, cutting clinical onboarding time 50% at Morgan Stanley and saving \$120K/month in chargebacks through a real-time fraud detection engine. Skilled in agent frameworks (LangGraph), the Model Context Protocol (MCP), production-grade RAG (LangChain, Pinecone, FAISS), and cloud-native ML delivery across AWS, GCP, and Kubernetes. Aiming to apply this AI/ML engineering expertise to accelerate intelligent, data-driven systems for a forward-thinking organization.

KEY ACHIEVEMENTS

- **50% reduction:** in clinical onboarding time and 38% fewer hallucinations via a LangGraph multi-agent RAG system at Morgan Stanley.
- **\$120K per month:** in chargebacks saved through a real-time fraud detection engine (Kafka, Spark Streaming, LangGraph) at sub-100ms latency.
- **92%+ accuracy:** with <2s P95 latency on production RAG pipelines processing 5M+ daily transaction events on AWS Bedrock.
- **52% reduction:** in post-deployment incidents and MTTR cut from 2.5 hours to under 30 minutes across 18 production ML services.
- **~1,200 hours per year:** saved and 25% retention improvement via RAG-based invoice classification and churn-prediction agents at Happiest Minds.
- **38% hallucination reduction, 42% token-cost cut:** and audit cycles trimmed from 14 to 4 days on a 500K+ document enterprise RAG system.

TECHNICAL SKILLS

- **AI/LLM & Agentic Systems:** RAG (LangChain, FAISS, Pinecone, ChromaDB, Weaviate), Agentic AI, LangGraph, Model Context Protocol (MCP), vLLM, Hugging Face (Transformers, PEFT), DeepSpeed, ONNX Runtime, Azure OpenAI Service, Azure AI Search, Prompt Flow, GPT-4, Claude 3.5, AWS Bedrock, Embedding Models, RAGAS, TruLens, Guardrails AI, NeMo Guardrails, Multi-Modal (Vision-Language) Models, Distributed Training (Horovod, Ray)
- **Languages & Scripting:** Python, SQL, TypeScript/JavaScript, Java, Scala, Bash, R
- **Machine Learning & Deep Learning:** TensorFlow, PyTorch, Scikit-learn, XGBoost, RLHF, PEFT/LoRA, Fine-Tuning, Anomaly Detection, Predictive Modeling, Statistical Hypothesis Testing, pandas/NumPy/SciPy, Time-Series Analysis, Causal Inference, Regression, Classification, Clustering, Feature Engineering, A/B Testing & Experiment Design, Model Evaluation (AUC-ROC, Precision/Recall), Jupyter Notebooks
- **Data Engineering & ETL:** Apache Spark, Apache Airflow, dbt, Snowflake, Kafka, Feast, Apache Iceberg, Delta Lake, MongoDB, Data Modeling (Kimball), Azure Databricks, Azure Data Factory, ADLS Gen2, Synapse Analytics, Redshift, BigQuery, Great Expectations, dbt Tests, Apache Flink, Kinesis Data Streams, T-SQL, PL/SQL
- **Cloud Platforms:** AWS (Bedrock, SageMaker, EKS, Kinesis), GCP (Vertex AI), Azure (Azure OpenAI, Azure ML, Azure Databricks, Azure Data Factory, ADLS Gen2, AKS)
- **Data Governance & Compliance:** Ataccama, Informatica, Collibra, RBAC, Metadata Management, Data Lineage, Data Quality Management, HIPAA
- **DevOps & MLOps:** Docker, Kubernetes, Terraform, Jenkins, CI/CD, GitLab, Git/GitHub, MLflow, Kubeflow, pytest & Unit Testing
- **Observability & Monitoring:** Grafana, Prometheus, Datadog, AWS CloudWatch, ELK Stack, Evidently AI, PagerDuty
- **BI & Visualization:** Tableau, Power BI, Plotly, Streamlit, SHAP Dashboards
- **APIs & Integration:** FastAPI, Flask, REST APIs, Jira/Confluence APIs, Microservices

PROFESSIONAL EXPERIENCE

Morgan Stanley

Jul 2025 - Present

AI and ML Engineer

New York, USA

- Designed and deployed a LangGraph multi-agent system with MCP-style tool-use patterns, integrating GPT-4 and HIPAA-compliant RAG pipelines (LangChain + FAISS over EHR data) — cut clinical onboarding time 50% and reduced hallucinations 38% across 10,000+ monthly queries.
- Built production RAG pipelines on AWS Bedrock (Pinecone + OpenAI + LangChain) for financial document retrieval and automated compliance reporting, achieving sub-2s P95 latency with 92%+ accuracy across 5M+ daily transaction events.
- Architected agentic AI dialog workflows with agent-tool integration via MCP patterns, instrumented with Datadog + Evidently AI for response accuracy, drift, and latency — achieving 99.5% SLA compliance; deployed via Kubernetes (EKS) + MLflow.
- Engineered ETL pipelines and feature engineering using Spark, Airflow, Snowflake, and dbt on GCP Vertex AI, reducing batch latency 35% across 50+ datasets (2TB per day) and managing data quality, validation, and drift detection across 12 production models.
- Integrated GPT-4 + LangGraph agents with real-time market data feeds via AWS Kinesis, processing 1M+ signals per day, and built SQL-backed analytics with Athena, improving risk-flagging precision 31% and reducing analyst review time 4 hours per day.

- Standardized model evaluation and debugging frameworks across 18 production ML services, reducing post-deployment incidents 52% and cutting MTTR from 2.5 hours to under 30 minutes.
- Implemented RAG evaluation and guardrail frameworks (RAGAS, TruLens, Guardrails AI) to monitor answer faithfulness and enforce output safety policies across production agent workflows, reducing unsafe or off-policy responses.
- Established Great Expectations data-quality checks and Git-based CI/CD workflows for the ML codebase, standardizing unit-test coverage (pytest) across pipelines and cutting data-related production defects.

Happiest Minds Technologies

Sep 2020 - Jun 2024

Machine Learning Engineer

Hyderabad, India

- Built LangGraph-orchestrated AI agents with tool-use patterns for churn prediction (Gradient Boosting + SHAP) and implemented RAG-based NLP invoice classification via Hugging Face, saving ~1,200 person-hours per year with a 25% retention improvement.
- Engineered payment default forecasting and anomaly detection models (XGBoost, TensorFlow) on AWS SageMaker, achieving 92% accuracy and 40% fewer false positives; deployed real-time churn APIs (Flask + Docker, AWS ECS) with CloudWatch dashboards.
- Designed ETL pipelines and data quality workflows on AWS Glue + SageMaker, orchestrated with Airflow across 8 ML services, reducing model degradation 45% and cutting time-to-production 30% via automated validation and rollback mechanisms.
- Led CI-CD pipeline setup with GitLab + Terraform for an ML platform migration to AWS SageMaker, completing 3 weeks ahead of schedule while supporting 20+ production models at 99.9% uptime and cutting deployment overhead 40%.
- Built automated A-B testing infrastructure using SageMaker Experiments, reducing manual evaluation cycles 65% and accelerating release cadence from bi-monthly to bi-weekly — contributing to a 12% YoY revenue uplift from improved model recommendations.
- Applied statistical hypothesis testing and time-series forecasting (pandas, NumPy, SciPy) to model demand and revenue trends, feeding results into cross-functional planning reviews.
- Migrated legacy reporting pipelines into Redshift and BigQuery data warehouses, standardizing data modeling (Kimball-style) and improving query performance for downstream analytics teams.
- Partnered with data governance stakeholders to apply RBAC and metadata management practices across ML feature stores, improving data lineage traceability and audit readiness.

EDUCATION

Saint Francis College

Brooklyn, NY

Master of Science, Computer Science

- GPA: 3.96/4

PROJECT HIGHLIGHTS

Enterprise Multi-Modal Agentic RAG System [Live Demo](#)

Python, LangGraph, LangChain, MCP Tool Patterns, GPT-4 and Claude 3.5, Pinecone, AWS Bedrock

- Designed a LangGraph multi-agent RAG pipeline with MCP-compatible tool-use, hybrid retrieval (dense + BM25), dynamic chunking, and OCR (Unstructured + Tesseract) across 500K+ documents — cut hallucinations 38%, token costs 42%, and audit cycles 70% (14 to 4 days) at sub-2.1s P95 latency.
- Scaled to 10,000+ concurrent enterprise users via EKS auto-scaling with 99.8% retrieval precision, reducing manual review effort 60% and saving ~3,500 person-hours per quarter.

Real-Time Fraud Detection with Full MLOps Pipeline [Live Demo](#)

Python, Kafka, Spark Streaming, LangGraph, Feast, Redis, MLflow, Kubernetes (EKS), Prometheus, Grafana

- Built an end-to-end MLOps pipeline with LangGraph AI agents, a Feast + Redis feature store (sub-5ms lookup), and ETL validation with auto-retrain on drift — reduced false-positive rate 35% and saved \$120K per month in chargebacks across 5M+ daily transactions at sub-100ms latency.
- Implemented Prometheus + Grafana alerting with PagerDuty, achieving MTTD under 90 seconds on model drift while processing \$2.4B+ monthly transaction volume at 99.97% availability.

Computer Vision QA Pipeline – Edge AI for Autonomous Inspection [Live Demo](#)

PyTorch, TensorRT, AWS S3 and EC2, GradCAM, Tableau

- Deployed YOLOv8 + a custom CNN at 60 FPS via TensorRT FP16 (55% memory reduction), cutting defect escape rate below 0.5% and saving \$180K per year.
- Built Grad-CAM + Tableau dashboards that cut false-alarm resolution time 50% and lifted QA trust scores from 62% to 94% across 12 edge devices.

CERTIFICATIONS

- GenAI Job Simulation — BCG X
- Data Analytics — Quantum
- Data Analytics — Deloitte
- Data Science — BCG X